Machine Learning for teens

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Spam detection email

- CountVectorizer
  - converts text into numbers so machine learning models can understand it.
    - It creates Bag of Words model
    - Each unique word in your dataset becomes a feature (column).
    - Each message (row) is represented by how many times each word appears.

Review of evaluation metrics for classification

- Accuracy alone might look high if spam is rare (e.g., 95% ham, 5% spam).
- Precision ensures users don’t get annoyed by false spam flags.
- Recall ensures spam doesn’t slip through.
- F1 gives a balanced view when both are important

Outlier detection and removal using **IQR**

- Outlier detection and removal using **IQR**
  - First you have to know percentile , we already talk about it
    - 25th percentile (Q1) → 25% of the data is less than or equal to this value.
    - 50th percentile (Median, Q2) → 50% of the data is less than or equal to this value.
    - 75th percentile (Q3) → 75% of the data is less than or equal to this value.
  - Now IQR is
    - IQR = Q3 − Q1
  - Define bounds
    - lower_bound = Q1 - 1.5 * IQR
    - upper_bound = Q3 + 1.5 * IQR
  - Remove outliers

Assignment

- Use Reviews.csv
  - Do some data preprocessing and analysis to prepare it for training.
- I want to use melb_data.csv
  - Use all outliers techniques(percentiles, z-score and IQR) to remove outliers